James Lee

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Education

Carnegie Mellon University Aug 2022 - May 2026

Bachelor of Science in Information Systems, Double Major in Computer Science

- Relevant Coursework: Data Structures & Algorithms, Database Design, Computer Vision, Probability Theory

Experience

Amazon May 2025 – Aug 2025

Software Engineer Intern *Bellevue, WA*

- Created RAG chatbot with S3 knowledge base and Claude API translating user stories to MVEL rules
- Designed validation sandbox backtesting MVEL logic on historical data; preventing false alerts pre-production
- Scaled rule deployment 4x (50→200+ rules); eliminating PM/engineer bottleneck on manual rule creation

Forge Rewards - Y Combinator May 2024 - Aug 2024

Software Engineer Intern *Palo Alto, CA*

- Scaled mobile loyalty platform helping small businesses retain customers and compete with large chains
- Built XGBoost uplift model preventing notification fatigue; DynamoDB backend with real-time point tracking
- Generated $100K+ merchant revenue via A/B-tested targeted campaigns improving customer purchase frequency

CMU Tartan Student Fund Jan 2023 - Aug 2024

Undergraduate Analyst *Pittsburgh, PA*

- Pitched Costco to committee after identifying membership fees as undervalued recurring revenue stream
- Built valuation model isolating membership revenue; recurring fees provided downside protection during volatility
- Position returned 89% vs S&P's 51% over 22 months; membership thesis proved durable through market cycles

LIKELION - Tech Entrepreneurship Club Sep 2023 - Oct 2023

Teaching Assistant *Pittsburgh, PA*

- Developed ML curriculum (PyTorch, CNNs, deployment) for 20+ students building ML-powered products
- Mentored winning hackathon team building RCNN/ResNet50 image classifier on Kaggle lunar crater dataset

Carnegie Mellon University Jun 2023 - July 2023

Quant Finance Research *Pittsburgh, PA*

- Built options pricing simulator comparing risk-neutral valuation vs real-world profitability using 3 binomial models
- Used CRR/Shreve/Drift-Adjusted pricing engines with Monte Carlo (1000+ paths); validating convergence
- Developed strike optimization algorithm via grid search over 500+ combinations; deployed interactive Streamlit interface

Projects

Jane Street Market Prediction (Kaggle)

- Built ensemble model combining XGBoost/autoencoder-MLP predicting profitable trades on 500 days of HFT data
- Leveraged XGBoost for non-linear feature interactions while autoencoder separated signal from noise
- Implemented transaction cost modeling and time-series cross-validation to prevent lookahead bias
- Reduced feature dimensionality from 130 to 26 features preserving 95% variance, accelerating model inference

Two Sigma Financial Market Prediction (Kaggle)

- Built LightGBM classifier predicting 10-day stock returns integrating price history and NLP news sentiment
- Selected gradient boosting over neural networks for efficient handling of sparse news coverage (30% missing)
- Implemented Bayesian hyperparameter optimization with Optuna for automated model tuning across 50+ trials
- Validated predictions using walk-forward backtesting preventing lookahead bias with proper temporal splits

Technical Skills

Languages: Python, SQL, R, JavaScript, TypeScript, MATLAB
ML/Data: PyTorch, scikit-learn, XGBoost, LightGBM, NumPy, Pandas, statsmodels
Cloud: AWS (Bedrock, S3, Lambda, SQS, SNS), Docker, GCP
Interests: Top 5 out of 97 in CMU Poker Club X Jump Trading 2025